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(54) **LENTIVIRUS INTEGRATION JUNCTION ANALYSIS**

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(71) Applicant: **Miltenyi Biotec B.V. & Co. KG**,
Bergisch Gladbach (DE)

(72) Inventors: **Robert Pinard**, Lowell, MA (US);
Seiyu Hosono, Stoneham, MA (US);
Reto Mueller, Bergisch Gladbach (DE);
Jinling Wang, Waltham, MA (US)

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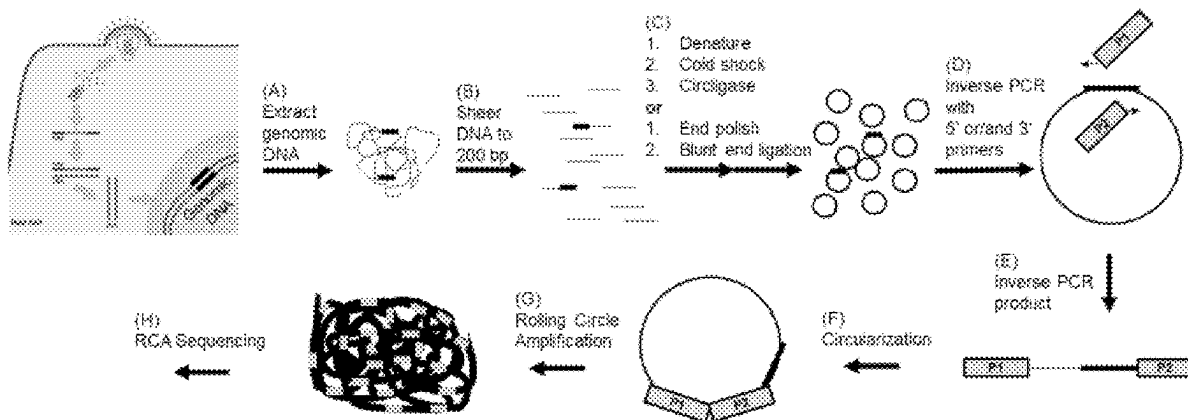
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(57)

ABSTRACT

The invention is directed to method to obtain the genomic location of a provirus sequence embedded by two LTR regions in a DNA strand

Specification includes a Sequence Listing.



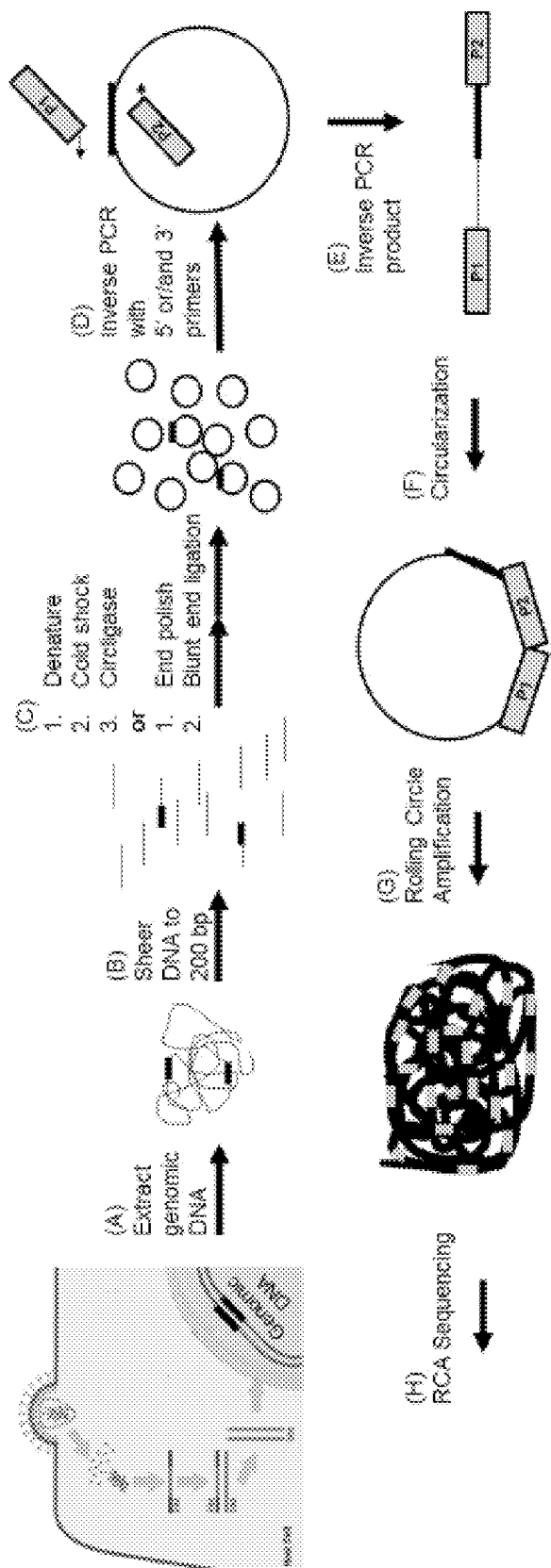


Fig. 1

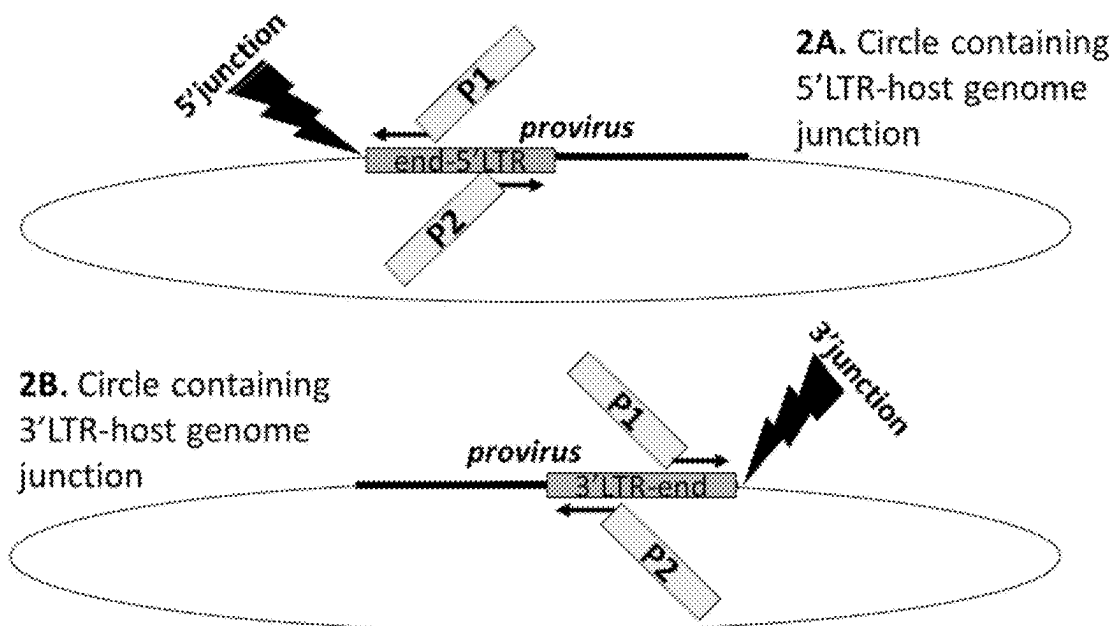


Fig. 2

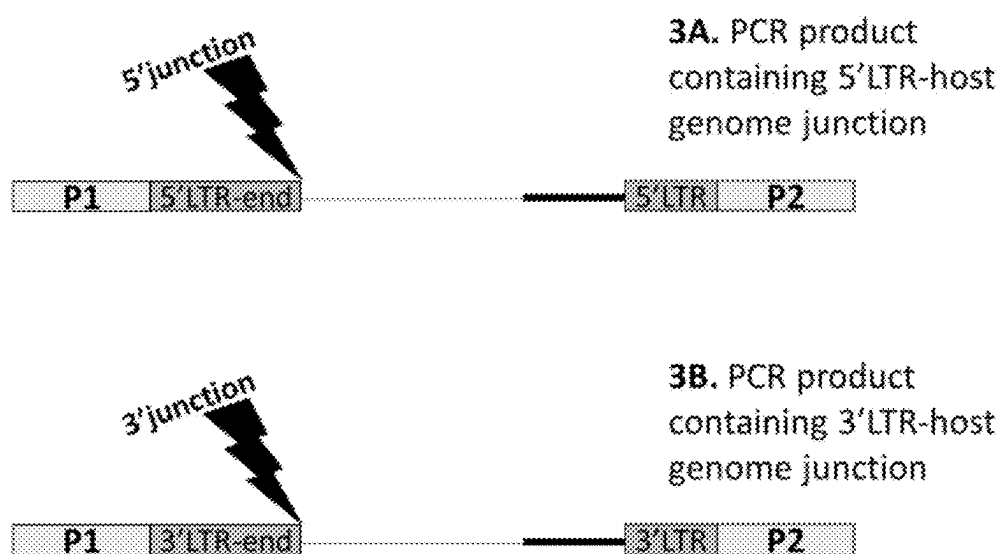
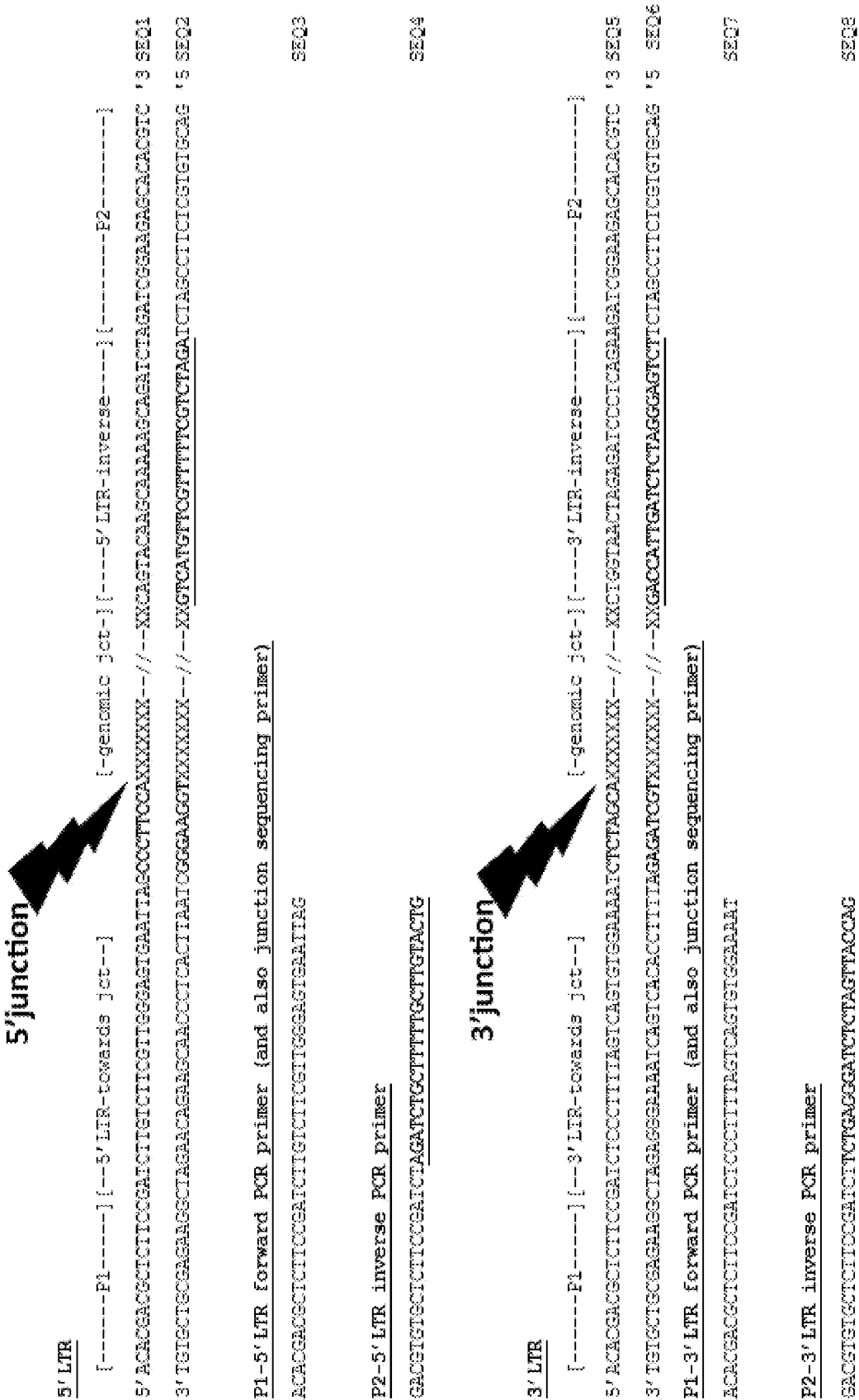


Fig. 3



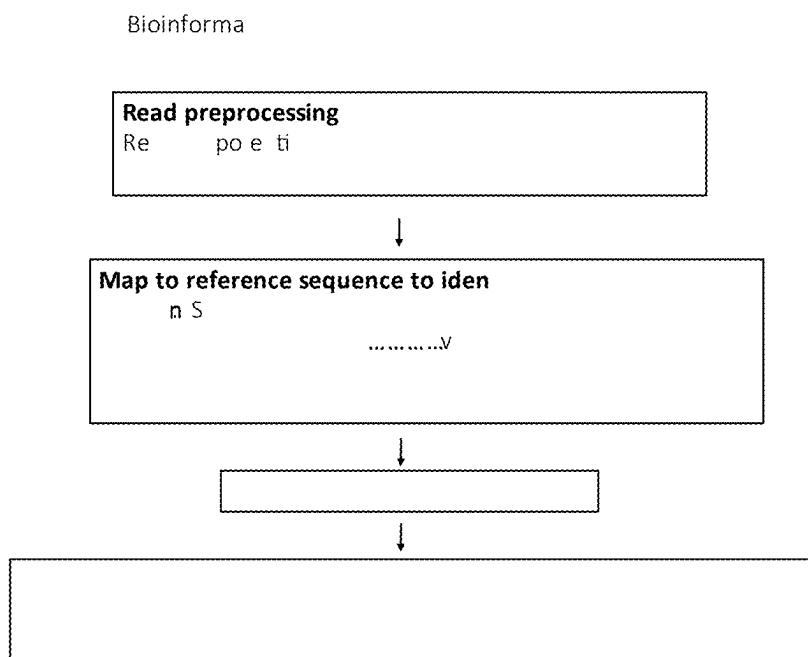


Fig 5

Fig 6

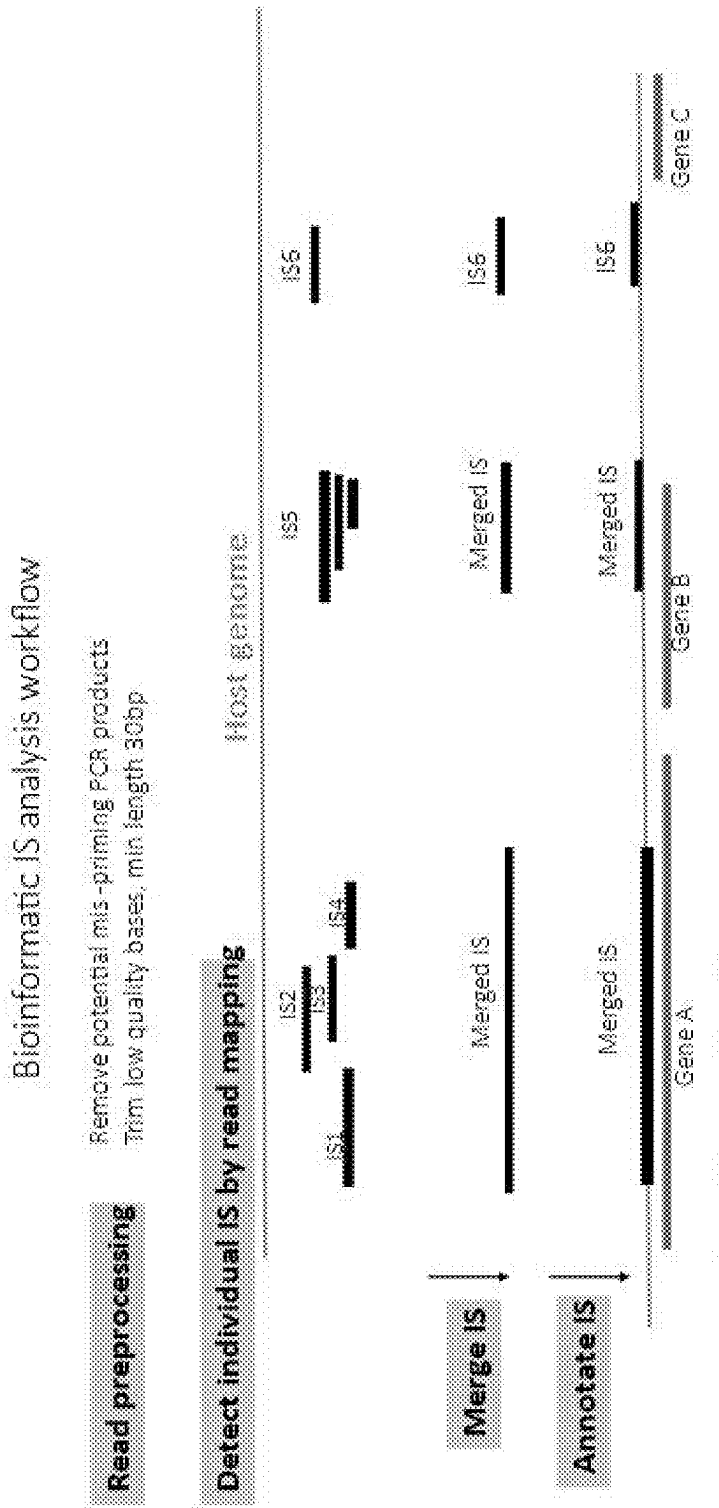


Fig 6

Using lentiviral mapping results to validate the IS analysis process
Reads mapped to 5' and 3' terminal region as expected

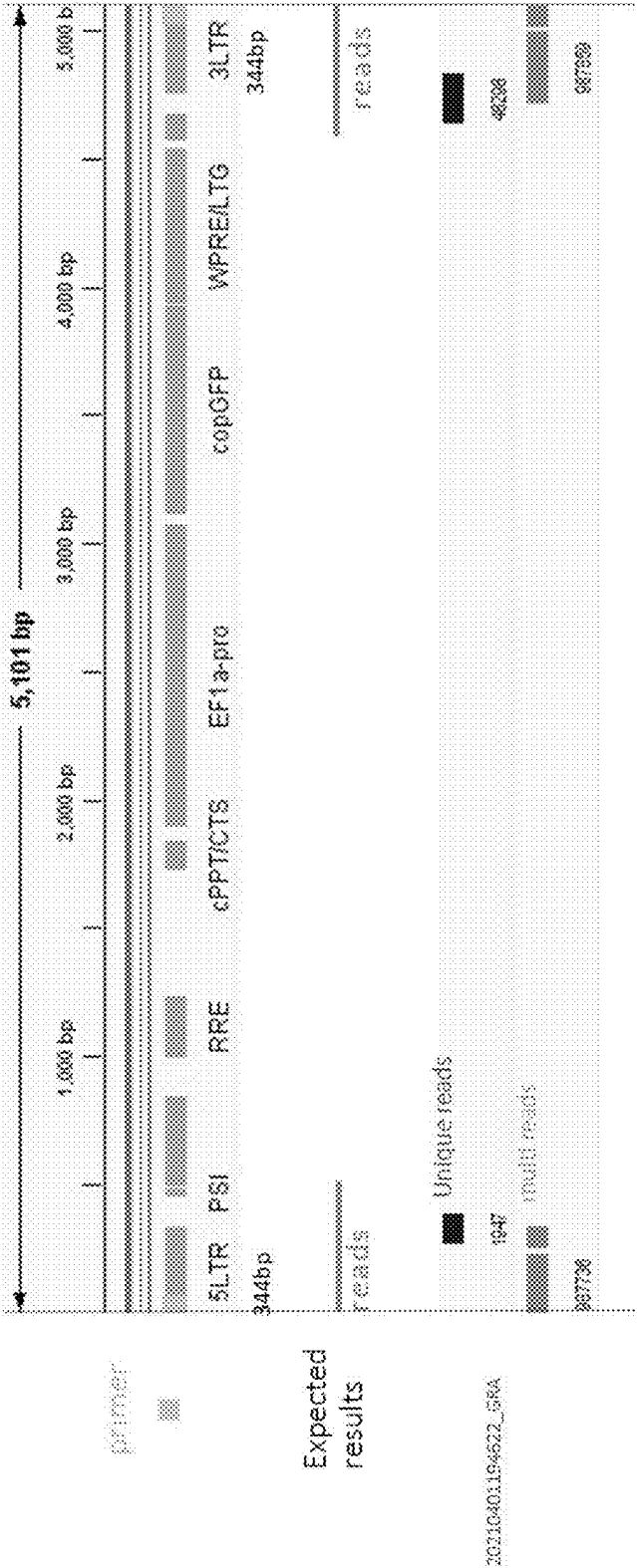


Fig 7

Unique IS distribution along host cell genome

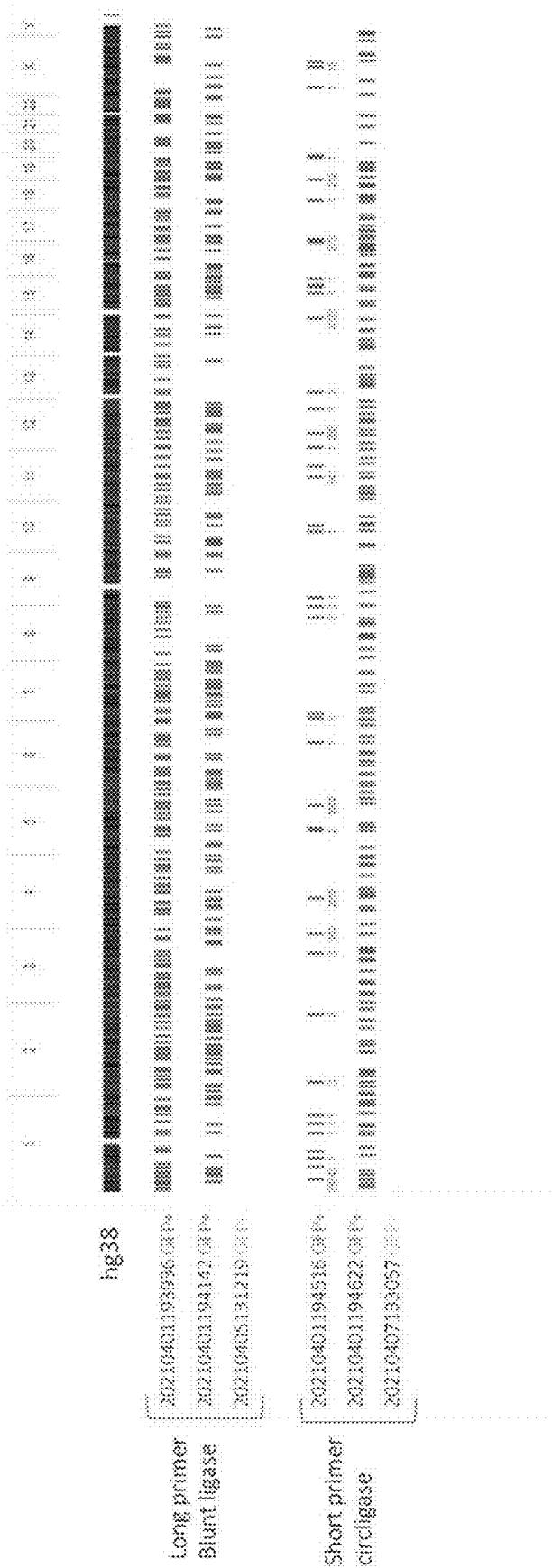


Fig. 8

Unique IS site summary

Group	fq	IS merge	IS IntersectGene	Uniq IntersectGene	IS outsideGene	Uniq closestGene	Total unig_gene
long primer, blunt ligase (GFP+)	20210401104507_38A	374	311	295	83	81	374
	20210401104507_38A	253	199	193	69	68	261
long primer, blunt ligase (GFP-)	20210405113145_38A	0	0	0	0	0	0
short primer, circligase (GFP+)	20210401104516_38A	46	30	30	17	17	47
	20210401104517_38A	240	150	145	99	98	243
short primer, circdigase (GFP-)	20210407113057_38A	0	0	0	0	0	0
sum		913	690	663	268	264	925

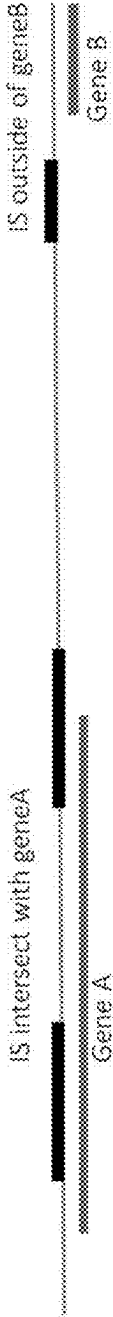


Fig 9

Example: unique IS report in each sample

group	file	chrom	is_chromStart	is_chromEnd	gene	distance
long primer: blunt ligase: GFP+	20210401194936_GRA	14	101977419	101977461	DYNCLH1	0
		19	1639808	1639847	TCF3	0
	20210401194142_GRA	20	33337536	33337571	BPIFB5P	7975
		12	94416096	94416165	CEP83	0
short primer: blunt ligase: GFP+	20210401194516_GRA	7	21468773	21468806	SP4	0
		11	93677764	93677840	CLP295	0
	20210401194622_GRA	17	18927750	18927780	PRPSAP2	0
		4	94237607	94237669	SMARCD1	0

Complete IS list was saved in excel worksheets
Subset genes can be retrieved such as TMB genes

Fig 10

- Measure the RCA generated by LTR-spec-
RCA and compare the copy number
rela
s g

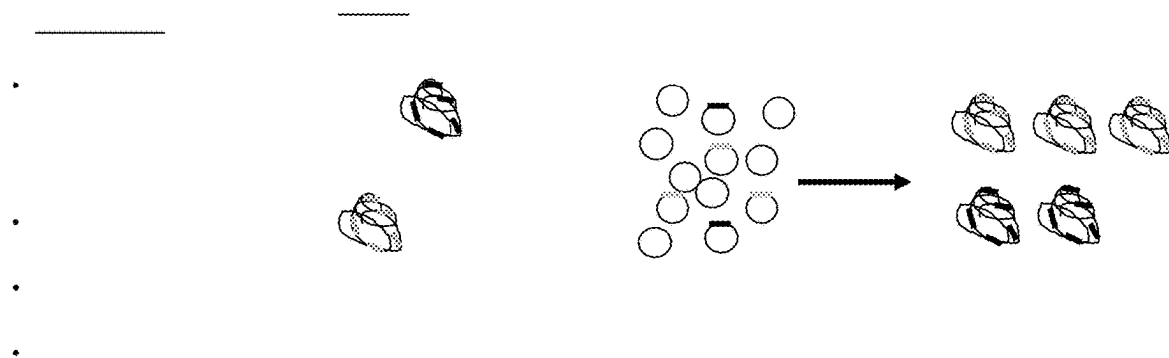


Fig. 11

LENTIVIRUS INTEGRATION JUNCTION ANALYSIS

BACKGROUND

[0001] Lentiviral vector delivery has advantages over other gene-therapy methods due to high-efficiency of infection of both dividing and non-dividing cells, long-term stable expression of a transgene, and low immunogenicity. Lentiviruses have been used to elicit an immune response against tumor antigens. However, like most current gene therapy experiments, establishment of a quick and accurate method of quality assessment of the virus delivery and infection is necessary.

[0002] Quick and efficient analysis of the location and the lentivirus provirus integration in the host human genome is essential for assessing the quality of viral transduction used for gene therapy.

[0003] This invention describes the method for assessing the location and optionally the copy number of lentiviral integration in the host human genome based on shearing the extracted genomic DNA followed by inverse PCR using a primer set from lentiviral LTR (Long terminal repeat) region and sequencing by Rolling Circle Amplification. The sequencing result is then analyzed by a customized bioinformatic algorithm to fish out the exact integration junction between the lentivirus provirus and the host genome.

SUMMARY

[0004] The goal of this invention is to provide a method to obtain the exact genomic integration location and optionally the copy number information of the transduced and integrated lentivirus (provirus) with a higher resolution than the known technologies.

[0005] The method involves PCR pre-amplifying a fragment of genomic DNA which carries the junction sequence of 5' and 3' LTR region and the host genomic DNA and rolling circle amplifying the PCR product.

OBJECT OF THE INVENTION

[0006] Object of the invention is a method to obtain the genomic location of a provirus sequence embedded by two LTR regions in a DNA strand characterized by the steps

[0007] a. fragmentation of the DNA strand into a plurality of strands having a length of 50 to 1000 bp thereby obtaining a mixture of strands comprising at least one LTR region of the provirus sequence and a junction region of the DNA strand having 20 to 100 bp and strands not comprising an LTR region

[0008] b. Converting the strands into circularized strands

[0009] c. Providing PCR primers P1 and P2 to the 3' and 5' ends of the at least one LTR region in the circularized strands

[0010] d. Multiplying the circularized strands provided with the PCR primers by inverse PCR, thereby obtaining a plurality of linear strands wherein the at least one LTR region and the junction regions are embedded by the PCR primers

[0011] e. Converting the linear strands with embedded LTR and junction regions into circularized and amplification of the circularized strands by RCA into colonies

[0012] f. Obtaining the sequence information of the colonies, thereby obtaining the sequence information of the junction regions

[0013] g. Aligning the sequence information of the junction regions with the sequence information of the DNA strand, thereby obtaining the genomic location of the provirus sequence.

[0014] The method according to the invention is especially suitable to obtain the genomic location of a provirus sequence in a DNA strand which is the human genome sequence of the human chromosomes.

[0015] The provirus sequence may origin from any kind of virus like a lentivirus. The term "provirus sequence" and "lentivirus sequence" are used interchangeable.

[0016] In addition to obtain the genomic location of a provirus sequence, the number of copies of the provirus sequence in the DNA strand can be obtained.

[0017] To this end, the sequence information of the DNA strand should be obtained either before or after the method. Since the human genome sequence is known, this information is accessible to the person skilled in the art. However, the sequence information of the DNA strand can also be obtained by the standard DNA sequencing methods.

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] FIG. 1 shows the general workflow of the method of the invention

[0019] FIG. 2 shows circularized strands with PCR primers P1 and P2 provided to the at least one LTR region (Step c)

[0020] FIG. 3 shows the linear strands multiplied by PCR obtained in step d)

[0021] FIG. 4 shows the exact DNA sequence of the construct of PCR primer used in inverse PCR reaction described in FIGS. 2 and 3.

[0022] FIGS. 5 and 6 show the bioinformatic workflow of aligning the information of the junction region with the sequence of the DNA strand (step g)

[0023] FIG. 7 shows the location of the lentiviral insertion site along the lentivirus genome. All reads mapped to 5' and 3' terminal region of the LTR region as expected in Lentivirus integrated samples.

[0024] FIG. 8 shows the unique insertion site distribution along host cell genome in an test experiment performed with Lentivirus integrated samples (GFP+).

[0025] FIG. 9 shows the summary of the proof-of-concept experiment from 2 different methods (long primer blunt ligase and short primer CircLigase) on positive control GFP+ and negative control GFP-samples. IS merge is the total number of samples that was found to be integrated within certain genes.

[0026] FIG. 10 shows the detailed result (from 4 individual sequencing datasets) from 2 different methods (long primer blunt ligase and short primer CircLigase) on positive control GFP+.

[0027] FIG. 11 shows how to determine the provirus (integrated lentivirus) copy number per cell.

DETAILED DESCRIPTION

[0028] The general method of the invention is shown in FIG. 1 which is now described in detail. The method is employed at cells transduced with a lentivirus. Such transduction processes are known the person skilled in the art.

[0029] First, genomic DNA is extracted from transduced cells (FIG. 1A).

[0030] Next, the genomic DNA is sheared either mechanically or enzymatically preferable to an average size of 100-300 base pairs (bp) like 200 bp. (FIG. 1C).

[0031] In a first embodiment of the invention, the plurality of the strands is denatured into a plurality of single stranded DNA strands which is then converted into a plurality of circularized single stranded DNA. This can be achieved by mechanical denaturing, cold shocking and single strand DNA ligation (e.g. by CircLigase) of the denatured single strand.

[0032] In a different approach, the sheared genomic DNA is end polished and blunt end ligated with T4 DNA Ligase (FIG. 1C). Accordingly, in a second embodiment of the invention, after step a), the plurality of the strands is provided with blunt ends by a ligase to obtain a plurality of double stranded DNA strands which is then converted into a plurality of circularized double stranded DNA, preferable by a T4 ligase.

[0033] Next, the circularized genomic DNA which contains LTR sequence are amplified with Polymerase Chain Reaction using inverse PCR primers with adapters with a fixed DNA sequence P1 and P2 which binds to the LTR regions of Lentivirus (FIGS. 1D and 1E).

[0034] Two sets of PCR primers may be used. One set is specific for 5'LTR and another set is specific for 3'LTR since 5'LTR and 3'LTR will not coexist in the same sheared DNA molecule. The directionality of the PCR primer for 5'LTR region is so that primer P1 is facing towards the 5' end of the 5'LTR (FIG. 2A).

[0035] In a third embodiment of the method, the circularized strands provided with the PCR primers comprises a mixture of first circularized strands having junction regions in 3' direction of the LTR region, and second circularized strands having junction regions in 5' direction of the LTR region, and wherein the first and second circularized strands are independently multiplied by inverse PCR thereby obtaining of linear strands having junction regions in 3' direction of the LTR region or linear strands having junction regions in 5' direction of the LTR region.

[0036] In a forth embodiment of the method, the PCR primers P1 and/or P2 are provided to the respective 3' and/or 5' ends of the at least one LTR region in the circularized strands such that PCR primer P1 is facing towards the 5' end of the 5'LTR or 3'LTR region and PCR primer P2 is facing towards the 3' end of the 5'LTR or 3'LTR region.

[0037] In this embodiment, the directionality of the PCR primer for 3'LTR region is so that primer P1 is facing towards the 3' end of the 3'LTR (FIG. 2B). That way both genomic insertion junctions of the provirus can be mapped simultaneously. The orientation of the 5'LTR PCR product (FIG. 3A) and 3'LTR PCR product (FIG. 3B) is depicted.

[0038] Preferable, step c) is performed by using a DNA splint bridge oligonucleotide which brings the P1 and P2 ends together and ligation by T4 DNA Ligase (FIG. 1F).

[0039] The circularized strand is then amplified by Rolling Circle Amplification (RCA) (FIG. 1G) into so called rolonies. Rolonies comprise a plurality of linear concatamers of the circularized strand.

[0040] The resulting RCA product (rolony) is then NGS sequenced using a 5'LTR junction sequencing primer which binds 8 bases upstream of the 5' end of LTR and the genomic junction, and a 3'LTR junction sequencing primer which

binds 8 bases upstream of the 3' end of LTR and the genomic junction so that one can distinguish between the two ends (FIG. 4).

By NGS sequencing of the LTR and genomic DNA junction can be analyzed using the analysis workflow described below.

[0041] Step g) of the method of the invention may be performed by mapping the sequence information of the junction regions to the sequence information of the DNA strand wherein regions of the DNA strand aligned with high confidence (MAPQ $\geq$ 30) with to the sequence information of the junction regions are designated as genomic location of the provirus sequence. Aligning may include merging overlapping sites, and annotation with intersecting genes, or the closest genes if no intersecting genes were identified.

EXAMPLE

[0042] This example describes the method for assessing the location and copy number of lentiviral integrations in the host human genome according to the invention.

[0043] The method involves PCR pre-amplifying a fragment of genomic DNA which carries the junction sequence of 5' and 3' LTR region and the host genomic DNA and rolling circle amplifying the PCR product.

Transduction/Multiplication Steps

[0044] The genomic DNA from a Lentivirus vector transduced SUP-TI cells (Human T lymphoblast cell line from ATCC) is extracted and used against a genomic DNA extracted from a non-transduced cells as negative control. By way of example, a lentiviral vector construct derived from HIV was used, which is a highly efficient vehicles for in vivo gene delivery. The construct carries a Green Fluorescence Protein (GFP) which exhibits green fluorescence when exposed to light in the blue to ultraviolet range which serves as a marker for positive transduction and integration of provirus into the host genome.

[0045] First, genomic DNA extracted from lentivirus transduced (GFP+ plus) and non-transduced (GFP- minus) SUP-TI cells (FIG. 1A).

[0046] Next, the genomic DNA samples were sheared mechanically by sonication (Covaris Ultrasonicator) by setting the desired size to 200 base pair (FIG. 1C). The concentration of sheared DNA was 200 nanogram in 130 μ L volume of 1 $\times$ Tris/EDTA (1.54 ng/ μ L).

[0047] Next, in one method, 50 nanogram of sheared genomic DNA was denatured at 95° C. for 10 min, cold shocked at 4° C. and was set on ice. Ligation of a single-stranded circle was performed with CircLigase at 60° C. for 60 min and inactivated at 80° C. for 10 min. The ligation reaction was cleaned up to remove uncircularized molecule by incubate with Exonuclease I and III at 37° C. for 60 min and purified with Spry Bead or size exclusion column.

[0048] Alternatively, the fragmented genomic DNA was end polished and blunt-end ligated with T4 DNA Ligase (FIG. 1C).

[0049] Next the circularized genomic DNA which contains LTR sequence is amplified with Polymerase Chain Reaction using inverse PCR primers with adapters with a fixed DNA sequence P1 and P2 which binds to the LTR regions of Lentivirus (FIGS. 1D and 1E).

[0050] Two sets of PCR primers are used. One set is specific for 5'LTR and another set is specific for 3'LTR since

5'LTR and 3'LTR will not coexist in the same sheared DNA molecule of 200 base pair size. The directionality of the PCR primer for 5'LTR region is so that primer P1 is facing towards the 5' end of the 5'LTR (FIG. 2A).

[0051] In contrast, the directionality of the PCR primer for 3'LTR region is so that primer P1 is facing towards the 3' end of the 3'LTR (FIG. 2B). That way both junctions of the provirus can be mapped simultaneously. The orientation of the 5'LTR PCR product (FIG. 3A) and 3'LTR PCR product (FIG. 3B) is depicted.

[0052] The PCR amplified product can then be circularized using the splint bridge oligonucleotide primer which brings the P1 and P2 ends together (FIG. 1F).

[0053] The circle is then amplified by Rolling Circle Amplification (RCA) (FIG. 1G).

[0054] The resulting RCA product is then NGS sequenced using a 5'LTR junction sequencing primer which binds 8 bases upstream of the 5' end of LTR and the genomic junction, and a 3'LTR junction sequencing primer which binds 8 bases upstream of the 3' end of LTR and the genomic junction so that one can distinguish between the 3' and 5' ends (FIG. 4).

[0055] The genomic DNA/LTR junctions can then be sequenced, and the junction region can then be defined against the genomic DNA reference sequence.

Bioinformatic Insertion Site Analysis Workflow is Performed to Determine Integrated Sites (Coordinates on the Chromosome)

[0056] FIGS. 5 and 6 show the bioinformatic workflow for analyzing the sequencing result of the integration site (IS).

[0057] Read preprocessing is performed by removing potential mis-priming of PCR products & adapters. Low quality bases are trimmed, min length 30 bases pairs.

[0058] Next, the sequence is mapped to human reference sequence to identify the IS.

[0059] The sequence is also mapped to lentiviral vector sequence in order to validate the IS analysis process.

[0060] Next overlapped IS are merged and IS are annotated.

[0061] At the end, a report is generated and visualized.

[0062] Unique IS are identified with high confidence MAP Q>=30 (mis-mapping probability=<0.001%).

[0063] Ambiguous IS are also identified (Reads mapped to multi-location equally well).

[0064] FIG. 6 shows the bioinformatic workflow for analyzing the sequencing result of the integration site (IS).

[0065] FIGS. 7 to 10 shows a sample result generated from a proof-of-concept experiment performed with Lentivirus integrated samples (GFP+ plus) and a negative control (GFP- minus).

[0066] FIG. 7 shows that all the lentiviral insertion site mapping resulted in reads mapped to 5' and 3' terminal region of the LTR region as expected in Lentivirus integrated samples (GFP+). The number of read in 3' and 5' end was very close to each other (987,736 vs 987,859) which shows that this method is performing on both 3' and 5' ends equally well. The fact that the reads only mapped to LTR and adjacent regions, but not virus internal region suggests that this method is performing well.

[0067] FIG. 8 shows unique insertion site distribution along host cell genome in Lentivirus integrated samples (GFP+). Long primer blunt ligation method and short primer circcligase method both showed broad distribution along the host chromosome in different GFP+ samples.

[0068] FIG. 9 shows the summary of the proof-of-concept experiment from 2 different methods (long primer blunt ligase and short primer circcligase) on positive control GFP+ and negative control GFP-samples. IS merge is the total number of samples that was found to be integrated within certain genes. IS intersectGene is the total number of samples that was found to intersect certain genes. Uniq intersectGene is the total number total number of unique genes that intersect with viral insertion site. IS outsideGene is the total number of samples that was found outside of any genes. Uniq closestGene is the total number of unique genes that are close to 'IS outsideGene'. Total uniq-gene is the total number of genes that intersect or closest to IS (viral insertion site).

[0069] FIG. 10 shows the detailed result (from 4 individual sequencing datasets) from 2 different methods (long primer blunt ligase and short primer circcligase) on positive control GFP+. It shows in which chromosome the lentivirus was integrated (chrom). It shows where exactly the sequencing reads are mapped (IS_chromStart and IS_chromEnd) on which genes. It also shows the distance between the integration site and a certain gene.

[0070] FIG. 11 shows how to determine the provirus copy number. Use a single copy internal control gene e.g.—CD3 as a reference to determine the relative copy number of the LTR-provirus copy. First, RCAs are generated from LTR specific iPCR and single copy gene specific (e.g.—CD3) RCA. Secondly, NGS sequencing is performed on both LTR region and CD3 gene and read number is determined. Since single copy gene (CD3) has 2 copies per cell (diploid), divide the totally ready by 2 (e.g.—10,000 read divide by 2 is 5,000 copies). Divide the number of reads from LTR by single copy gene CD3 (15,000/5,000=3). There are 3 copies of provirus per cell. Alternatively, one can determine the copy number using a quantitative PCR for reference gene as well as LTR region of provirus.

SEQUENCE LISTING

```
Sequence total quantity: 8
SEQ ID NO: 1          moltype = DNA length = 94
FEATURE               Location/Qualifiers
source                1..94
                     mol_type = genomic DNA
                     organism = unidentified

SEQUENCE: 1
acacgacgct cttccgatct tgtcttcggt gggagtgaat tagcccttcc acagtacaag 60
caaaaagcag atctagatcg gaagagcaca cgct                                94
```

-continued

SEQ ID NO: 2	moltype = DNA length = 94	
FEATURE	Location/Qualifiers	
source	1..94	
	mol_type = genomic DNA	
	organism = unidentified	
SEQUENCE: 2		
tgtgctgcga gaaggctaga acagaagcaa ccctcactta atcgggaagg tgatcatgttc	60	
gtttttcgtc tagatctagc cttctcgtgt gcag	94	
SEQ ID NO: 3	moltype = DNA length = 43	
FEATURE	Location/Qualifiers	
source	1..43	
	mol_type = genomic DNA	
	organism = unidentified	
SEQUENCE: 3		
acacgacgct cttccgatct tgtcttcggt gggagtgaat tag	43	
SEQ ID NO: 4	moltype = DNA length = 43	
FEATURE	Location/Qualifiers	
source	1..43	
	mol_type = genomic DNA	
	organism = unidentified	
SEQUENCE: 4		
gacgtgtgct cttccgatct agatctgctt ttgcttgta ctg	43	
SEQ ID NO: 5	moltype = DNA length = 94	
FEATURE	Location/Qualifiers	
source	1..94	
	mol_type = genomic DNA	
	organism = unidentified	
SEQUENCE: 5		
acacgacgct cttccgatct cctttttagt cagtgtggaa aatctctagc actggtaact	60	
agagatccct cagaagatcg gaagagcaca cgtc	94	
SEQ ID NO: 6	moltype = DNA length = 94	
FEATURE	Location/Qualifiers	
source	1..94	
	mol_type = genomic DNA	
	organism = unidentified	
SEQUENCE: 6		
tgtgctgcga gaaggctaga gggaaaatca gtcacacctt ttagagatcg tgaccattga	60	
tctctaggga gtcttctagc cttctcgtgt gcag	94	
SEQ ID NO: 7	moltype = DNA length = 43	
FEATURE	Location/Qualifiers	
source	1..43	
	mol_type = genomic DNA	
	organism = unidentified	
SEQUENCE: 7		
acacgacgct cttccgatct cctttttagt cagtgtggaa aat	43	
SEQ ID NO: 8	moltype = DNA length = 43	
FEATURE	Location/Qualifiers	
source	1..43	
	mol_type = genomic DNA	
	organism = unidentified	
SEQUENCE: 8		
gacgtgtgct cttccgatct tctgagggat ctctagttac cag	43	

1. A method to obtain the genomic location of a provirus sequence embedded by two LTR regions in a DNA strand characterized by the steps

- a. fragmentation of the DNA strand into a plurality of strands having a length of 50 to 1000 bp thereby obtaining a mixture of strands comprising at least one LTR region of the provirus sequence and a junction region of the DNA strand having 20 to 100 bp and strands not comprising an LTR region
- b. Converting the strands into circularized strands
- c. Providing PCR primers P1 and P2 to the 3' and 5' ends of the at least one LTR region in the circularized strands

d. Multiplying the circularized strands provided with the PCR primers by inverse PCR, thereby obtaining a plurality of linear strands wherein the at least one LTR region and the junction regions are embedded by the PCR primers

e. Converting the linear strands with embedded LTR and junction regions into circularized and amplification of the circularized strands by RCA into colonies

f. Obtaining the sequence information of the colonies, thereby obtaining the sequence information of the junction regions

- g. Aligning the sequence information of the junction regions with the sequence information of the DNA strand, thereby obtaining the genomic location of the provirus sequence.
2. The method according to claim 1 characterized in that after step a), the plurality of the strands is denatured into a plurality of single stranded DNA strands which is then converted into a plurality of circularized single stranded DNA.
3. The method according to claim 1 characterized in that after step a), the plurality of the strands is provided with blunt ends by a ligase to obtain a plurality of double stranded DNA strands which is then converted into a plurality of circularized double stranded DNA.
4. The method according to claim 1 characterized in that step e) is performed by using a DNA splint bridge oligonucleotide which brings the P1 and P2 ends together and ligation by T4 DNA Ligase.
5. The method according to claim 1 characterized in that the circularized strands provided with the PCR primers comprises a mixture of first circularized strands having junction regions in 3' direction of the LTR region, and second circularized strands having junction regions in 5' direction of the LTR region, and wherein the first and second

circularized strands are independently multiplied by inverse PCR thereby obtaining of linear strands having junction regions in 3' direction of the LTR region or linear strands having junction regions in 5' direction of the LTR region.

6. The method according to claim 1 characterized in that the PCR primers P1 and/or P2 are provided to the respective 3' and/or 5' ends of the at least one LTR region in the circularized strands such that PCR primer P1 is facing towards the 5' end of the 5'LTR or 3'LTR region and PCR primer P2 is facing towards the 3' end of the 5'LTR or 3'LTR region.

7. The method according to claim 1 characterized in that the number of copies of the provirus sequence in the DNA strand is obtained.

8. The method according to claim 1 characterized in obtaining the sequence information of the DNA strand.

9. The method according to claim 1 characterized in that step g) is performed by mapping the sequence information of the junction regions to the sequence information of the DNA strand wherein regions of the DNA strand aligned with high confidence ($\text{MAPQ} \geq 30$) with to the sequence information of the junction regions are designated as genomic location of the provirus sequence.

* * * * *